

Create, Mount And Unmount .img Files In Ubuntu

📅 Oct 22, 2017

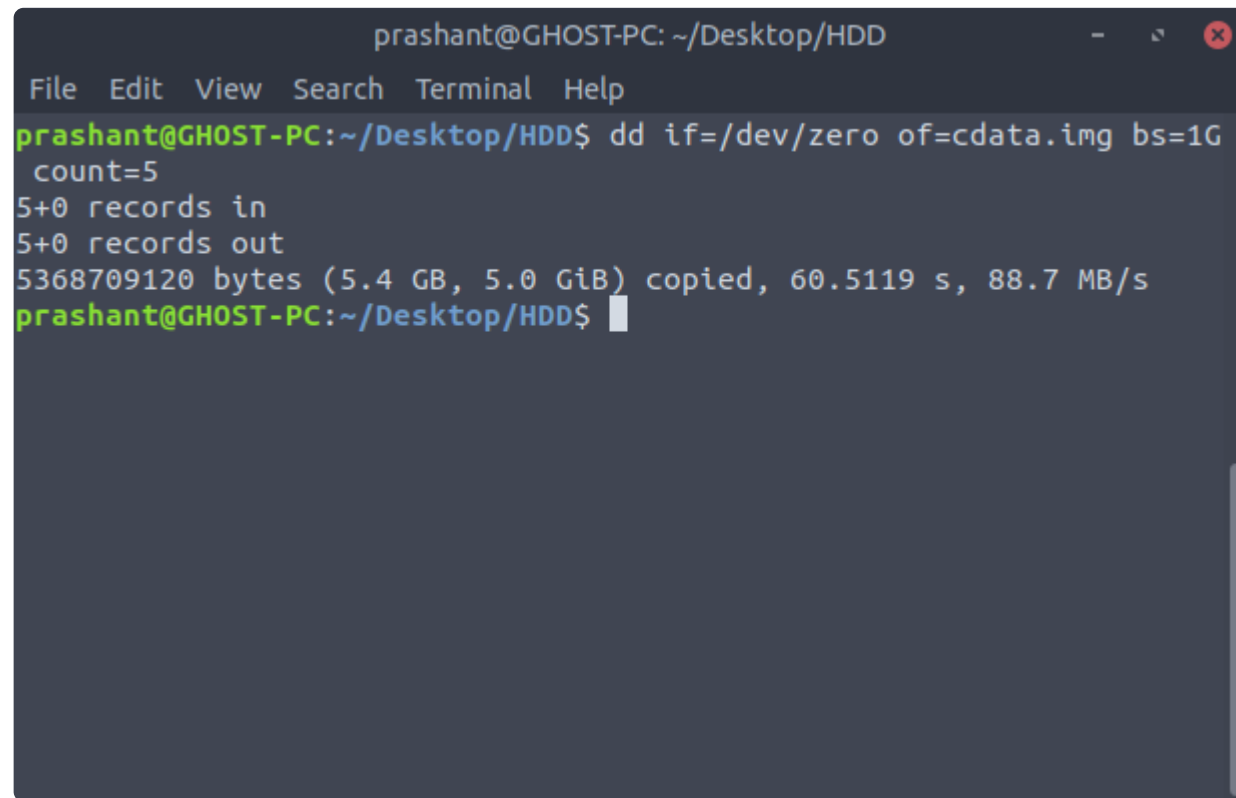
UBUNTU

Recently I moved from Windows to Ubuntu on one of my laptops and since then I was trying to look for the way to create a VHDs or some sort of image files that I can use as a container. So I looked up for the solution and found a working solution scattered though places. So here I am putting it in a blog post to have all it in one place.

First we need to create an image file (.img) by using the `dd` command. The `dd` command is used to copy and convert the file. The reason it is not called `cc` is because it is already being used by C compiler. Here goes the command.

```
1 | $ dd if=/dev/zero of=cdata.img bs=1G count=5
```

Note: The command execution time depends on the size of the image you have opted out for.



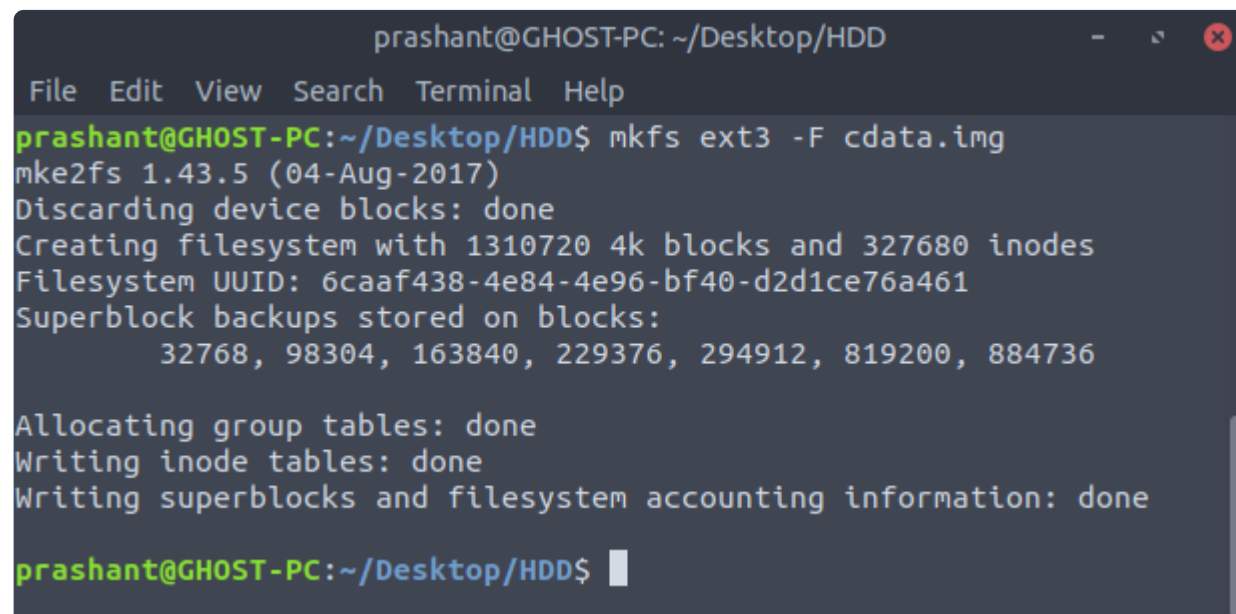
```
prashant@GHOST-PC: ~/Desktop/HDD
File Edit View Search Terminal Help
prashant@GHOST-PC:~/Desktop/HDD$ dd if=/dev/zero of=cdata.img bs=1G
count=5
5+0 records in
5+0 records out
5368709120 bytes (5.4 GB, 5.0 GiB) copied, 60.5119 s, 88.7 MB/s
prashant@GHOST-PC:~/Desktop/HDD$
```

The `if` param stands for `input file` to refer the input file. As `if`, `of` stands for `output file` and this will create the `.img` file with the name you specify. The third param `bs` stands for `block size`. This will let you set the size of the image file. The `G` in the above command stands for `GigaByte` (GB). You can set the image size by specifying the block size. In this case `G` stands for GigaBytes (GB), `K` for KiloBytes (KB), `T` for TeraBytes (TB), `P` for Petabytes (PB) and so on. Here I am creating a 5 blocks of 1 GB each for the image file name `cdata`.

This command will create the `.img` file but you cannot mount this file as of yet. To do that you have to format the image so that it can have a file system. To do this make use of `mkfs` command.

```
1 | $ mkfs ext3 -F cdata.img
```

I am creating an `ext3` filesystem on my image file. `-F` is to force the operation. Here is the output of the command.



```
prashant@GHOST-PC: ~/Desktop/HDD
File Edit View Search Terminal Help
prashant@GHOST-PC:~/Desktop/HDD$ mkfs ext3 -F cdata.img
mke2fs 1.43.5 (04-Aug-2017)
Discarding device blocks: done
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 6caaf438-4e84-4e96-bf40-d2d1ce76a461
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

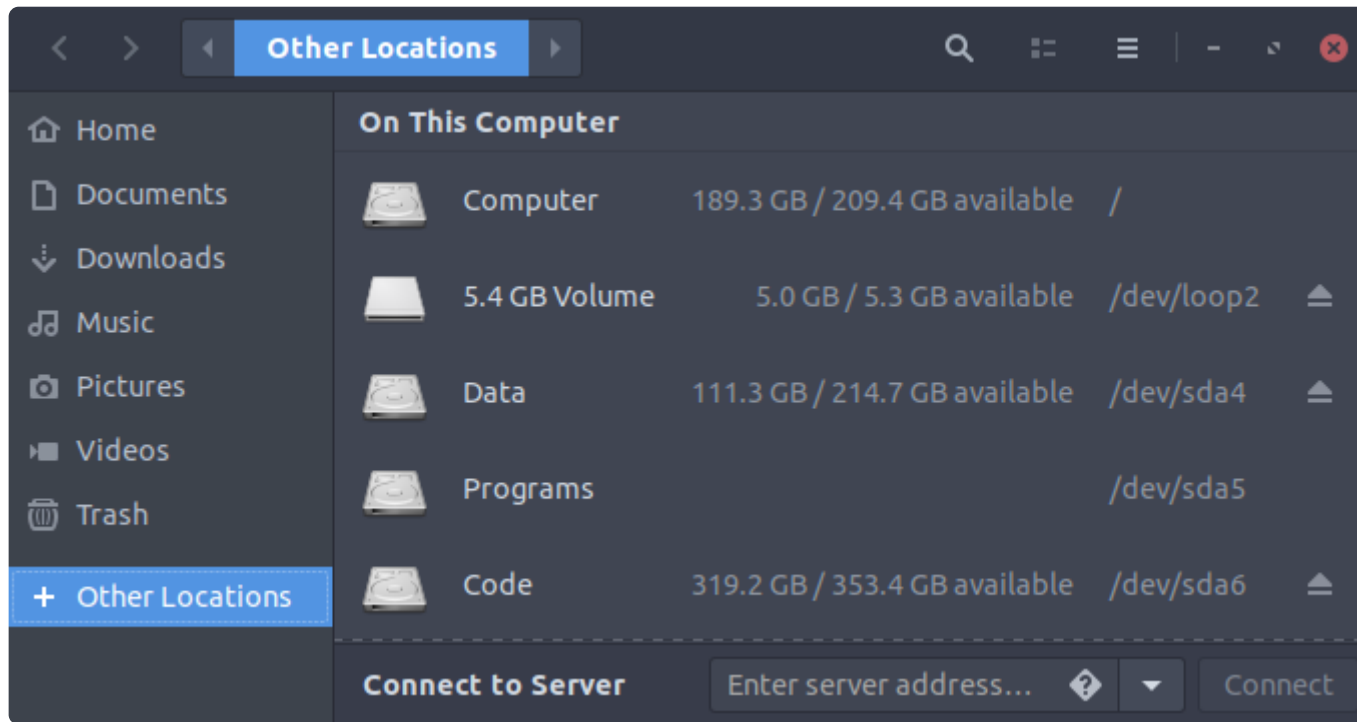
Allocating group tables: done
Writing inode tables: done
Writing superblocks and filesystem accounting information: done

prashant@GHOST-PC:~/Desktop/HDD$
```

Now we can mount the image file by using the `mount` command:

```
1 | $ sudo mount -o loop cdata.img ~/Desktop/HDD/
```

With the success of the above command, you can see the image mounted on your desktop. You can unmount the image by clicking the eject or unmount button as shown in the below screenshot or you can execute the `umount` command to do that.



Unmounting the image by using command line.

```
1 | $ sudo umount ~/Desktop/HDD
```